Module 2: Web Development

Deploy websites using frameworks and platforms.

Estimated Time: 3 hours

NOTE: When counting from 1 to 5, we go in order: 1, 2, 3, 4, 5. Similarly, when following a step-by-step process, it's important to go through each step in sequence without skipping around. If you find yourself stuck at any point, don't hesitate to reach out to your instructor for help and support.

If a text is in orange, then it is a link. A blue text should be typed **as is**. Then all red texts should be replaced by the expected value. Just ensure that you typed in **without misspellings**.

Learning Task 5 – Deploying Frontend Web Apps

In earlier days of web development, the backend and frontend web development is tightly integrated together. The popular tech stack, XAMPP (or W/LAMPP), was heavily used in various deployment environments.

However, most PHP web systems are often deployed on a **system unit** configured as a server that is accessible within the local area network. As the users becomes global, this local deployment is no longer sufficient even with **vertical scaling**—or the increase of resources (CPU, RAM, storage). Enterprise solutions often deploy similar systems in **cloud-hosting providers** which require developers to become “network administrators” as well. Consequently, **content management systems** such as Joomla and WordPress.org initiated the growth of frontend and low-code web development.

As the computing landscape continuously evolves, backend and frontend web development separates even further. It allowed the emergence of diverse development configurations, environments, and specialized roles to also cater with the modern IT needs. This flexibility is also called **serverless**, which means developers are let go of the technical needs of configuring physical servers. This is where **scalability** is almost automatic depending on the subscription plans. These innovations brought various tiers of web development which also requires different levels of broadness and depth of technical expertise.

The frontend is built with the client-side logic and the user interface while interacting with the backend through an Application Programming Interface (API). Modern backend services include authentication, database, hosting, storage, and other **cloud-based services**. Different cloud service providers typically offers an umbrella of services called ***Anything-as-a-Service*** (XaaS) but can be further categorized based on the configurability and flexibility of their services. However, a service

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provide may also offer multiple services for all developer needs which result to a vendor lock-in where switching to another vendor would involve significant cost and time.

Service	Providers
Infrastructure-as-a-Service (IaaS)	AWS, Azure, DigitalOcean, GCP
Platform-as-a-Service (PaaS)	Heroku, Netlify, Render, Vercel
Software-as-a-Service (SaaS)	Google Workspace, Microsoft 365

Cloud-based services provides both vertical scaling and **horizontal scaling**. Horizontal scaling is more complex to implement and manage but is better at **fault-tolerance** especially with high-traffic projects. **IaaS** demands the most advanced technical skills but offers the greatest flexibility in return. **PaaS** relieves developers from the complexity of configuring servers by offering budget-wise compute resources and other add-on services but typically limits the overall control. On the other hand, **Backend-as-a-Service** (BaaS) and **SaaS** provide solutions for database, storage, and other services that ensure the availability and reliability of the data generated and transmitted by users around the world. A common misconception tells that an **entry-level software engineer** is required to become an expert of these tech stacks and must be able glue these together to create a working system, but this is already another story.

Before we continue, make sure you are already acquainted with the technologies we used in earlier learning tasks. Otherwise, use the previous learning tasks as your guide as you go with the procedures in this document.

Vercel

If you crawl the internet, there is a never-ending list of cloud service providers available. Although most companies prefer big names, such as AWS, GCP, Microsoft Azure, and the likes, but the **barrier to entry** and the **cost of adoption** is high, which may not be intuitive for a new programmer. Other things to consider is the annual payments to analytics, domains, TLS certificates, and other services related to scaling, security, and maintenance to ensure uptime and performance. Learning these technologies require stronger foundational knowledge and skills and a financially-subsidized learning environment that immerse learners with real-world projects to ensure clear direction.

An alternative is to leverage hobby tiers (free) which does not hinder young developers without supported debit and credit cards. Mind you, a pay-wall is also a reality in testing the waters of software development. Although this may be limiting to most students, this is no longer applicable when working in tech companies.

1. Go to the LT5 Dashboard spreadsheet.
    - Improve the clarity and conciseness of your project description.
2. Launch your preferred web browser.

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   - Go to Vercel (vercel.com) website.
   - Create an account.
3. Go to **Account Settings** > **Authentication** link.
   - Link your GitHub account.
4. Go back to your Vercel dashboard.
   - Find and select the **Add New Project** option. (Note: the interface may differ.)
5. In the "**Import Git Repository**" container:
   - Ensure that your GitHub account is selected.
   - Type your repository name.
   - If not found, select "**Configure GitHub App**" button.
6. In the popup browser:
   - Authorize Vercel with your GitHub account.
   - Scroll down to "**Repository access**" setting.
   - Check that "**Only select repositories**" radio button is selected.
   - Search and select your target project repository.
   - Click the "Save" button.
7. In the "**Import Git Repository**" container:
   - In the target project, click the "**Import**" button.
8. In the "**New Project**" page:
   - Check if the project name is allowed.
   - If not allowed, append "-dev" to the project name.
   - Click the "**Deploy**" button.
   - Wait until it is completed.
9. In the "**Congratulations**" page:
   - The page will display your next steps which you may explore in your future projects.
   - Update the spreadsheet with your new website URL.

Frontend Web Development

Although there are multiple CSS frameworks available, let us start with the basics first. Cascading Style Sheets is a document used to define the presentation and style of HTML documents. Luckily, there is no need to install anything to start with CSS other than any modern browser. A simple text editor, such as Notepad++, is sufficient for most web design tasks. However, if your PC is able to handle VS Code or other IDEs, feel free to use it.

Note: In this section, you may use VS Code or your favorite IDE (recommended). When using VS Code, install the "Vue - Official" extension to avoid configuration issues. Optionally, you may still use Notepad++ as your text editor.

1. Open the Command Prompt:

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   a. Check the version: `node -v`
      i. Update NPM to latest: `npm install -g npm@latest`
      ii. If not yet installed, **reinstall** on your device.
2. Launch a web browser.
   a. Go to the LT5 Dashboard spreadsheet, update your deployment details.
3. Go back to Command Prompt.
   a. Clone the agreed repository locally.
   b. Navigate inside the repository.
      i. Install project dependencies: `corepack yarn install`
      ii. Create a "docs" folder.
         1. Navigate inside the "docs" folder.
            a. Create the "use-cases.txt" file.
            b. Create the "terms-of-service.txt" file.
            c. Create the "privacy-policy.txt" file.
      iii. Navigate upwards to the root of the project.
4. Go to **ChatGPT** or similar services.
   a. Submit a prompt in the format below.

Prompt 1
Create the use cases for a Minimum Viable Product (MVP) of "project description".

   b. Copy the **Use Case Scenarios** and paste into the "use-cases.txt" file.
   c. Save all changes.
5. In **Command Prompt**, push the changes with the commit message:

Command Prompt
```
git add "docs/use-cases.txt"
git commit -m "[Docs] Add `use-cases.txt` file"
git push -u origin main
```

6. Go back to the conversation, submit another prompt in the format below.

Prompt 2
Based on this use cases, draft a concise but explicit Terms of Service applying GDPR and the Data Privacy Act of 2012.

   a. Copy the **Terms of Service** and paste into the "terms-of-service.txt" file.
   b. Save all changes.
   c. Push the changes.
7. Go back to the conversation, submit another prompt in the format below.

Prompt 3
Based on this use cases, draft a concise but explicit Privacy Policy applying GDPR and the Data Privacy Act of 2012.

   a. Copy the **Privacy Policy** and paste into the "privacy-policy.txt" file.
   b. Save all changes.
   c. Push the changes.

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Web Design

The need for visually appealing and functional web interfaces became essential. Web design evolved beyond static pages, allowing for interactive and responsive experiences. While modern CSS frameworks simplify styling, building custom HTML and CSS first is a good training ground for understanding the fundamentals of layout, styling, and responsiveness. Mastering these core technologies ensures better control over designs before relying on frameworks. Let's start crafting custom web designs!

Note: Once you get gist of web design, you are free to customize the look and feel of the website that you are contributing into.

1. Go to **Google Fonts** (fonts.google.com/) website.
    a. In the preview section, type in the **codename** (Website Name) of the project.
        i. Choose one (1) font that fits the **Website Name**. (It can be classy or fancy)
        ii. Click the "**Get font**" button.
    b. Click the "**Fonts**" option in the left sidebar.
        i. Select the "**Reset all**" button.
        ii. Get one more font under the "Feeling – *Business*" and "Sans serif – *All*" category.
        iii. Check if "**2 font families selected**" is shown.
        iv. Click the "**Get embed code**" button.
        v. Copy the font (https://fonts.googleapis.com/...) URL.
2. In the succeeding snippets, replace "**bh**" with the short-code of the project codename.
3. In **Command Prompt**:
    a. Navigate to the root of the project directory.
    b. Delete all "components/*.vue" files.
    c. Navigate to the **public** folder.
        i. Create the "custom.css" file.
    d. Modify the CSS file with the following content.

custom.css

```
/* (c) 2025 Full Name */

* {
    box-sizing: border-box;
}

body {
      font-family: "Font Name #2", Arial, Helvetica, sans-serif;
      background-color: #f2f4f7;
}

/* navigation bar */
```


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```
/* main contents */

/* copyright */
```

e. Save all changes.
f. Push the changes.

3. In **Command Prompt**:
   a. Navigate upwards to the root of the project.
   b. Install package: `corepack yarn add "@vueuse/head"`
   c. Edit the "`pages/index.vue`" files.

index.vue

```
<script lang="ts">></script>

<template>
    <client-only>
        <!-- <NavBar /> -->
        <div id="bhMain" class="bh-main"></div>
        <!-- <Cookies /> -->
        <!-- <Copyright /> -->
    </client-only>
</template>

<script setup lang="ts">

import { useSeoMeta, useHead } from '@vueuse/head';

const title = "Code Name | Home";
const description = "Project description";

useSeoMeta({
  title: () => title,
  description: () => description,
  charset: "utf-8",
  viewport: "width=device-width, initial-scale=1.0"
});

useHead({
  link: [
    {rel: 'icon', type: 'image/png', href: '/logo.png'},
    {rel: 'stylesheet', href: '/reset.css'},
    {rel: 'stylesheet', href: '/custom.css'} ,
    {rel: 'preconnect', href: 'https://fonts.googleapis.com'},
    {rel: 'preconnect', href: 'https://fonts.gstatic.com', crossorigin: ''}
    {rel: 'stylesheet', href: 'fonts-url'}

  ]
});
```


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```
</script>

<style scoped></style>

<style></style>
```

d. Save all changes.
e. In **Command Prompt**, push the changes.
f. Repeat **Step 3** on the remaining "`pages/*.vue`" files. (Exclude `<Cookies />` component)

Color Palette

In branding, a careful case study is often conducted. It involves various factors such as the core values of the company, its mission and vision, and also its history—among others. However, a shortcut is to choose five colors, namely: **primary**, **secondary**, tertiary, neutral, and a neutral variant (See Material3 color schemes). However, a two-color scheme is sufficient for most applications.

1. Launch a web browser.
2. Go to **MaterialIU** (materialui.co/colors) website.
   a. Alternatively, got to **ColorHunt** (https://colorhunt.co/palettes/) for inspirations.
3. Choose the primary and secondary colors that suit the project.
   a. Take note of the **hex color codes** for the primary and secondary colors.
4. Create and edit the "`components/NavBar.vue`" file.

NavBar.vue

```
<template>
  <client-only>
    <header class="bh-navbar">
      <nav class="bh-nav-left">
          <a href="/" class="bh-nav-home">
            <img src="/logo.png" width=20 height=20 class="bh-nav-logo">
            <span>Bleedingheart</span>
          </a>
      </nav>
      <nav class="bh-nav-right">
        <ul>
          <li><a href="#" class="bh-nav-link">Feature #1</a></li>
          <li><a href="#" class="bh-nav-link">Feature #2</a></li>
          <li><a href="#" class="bh-nav-link">Feature #3</a></li>
        </ul>
      </nav>
    </header>
  </client-only>
</template>
```

   a. Replace the "Feature #X" with the three (3) main features as shown in the use cases.
   b. Save all changes.
5. Open the "`public/custom.css`" file.

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a. Append the following CSS rule sets under the `/* navigation bar */` comment.

custom.css

```
/* navigation bar */

.bh-navbar {
	position: fixed;
	left: 0;
	top: 0;
	width: 100%;
	height: 50px;
	background-color: #f0f0f0;
	box-shadow: 0 4px 6px rgba(0, 0, 0, 0.3);
	overflow: hidden;
	z-index: 7000;
}

.bh-nav-left {
	margin-left: 5%;
	display: inline-flex;
	width: fit-content;
	float: left;
}

.bh-nav-home {
	margin: 5px;
	padding: 10px 8px;
	display: flex;
	align-items: center;
	font-family: "Font Name #1", Arial, serif;
	font-size: 1.2em;
	text-decoration: none;
	border-radius: 8px;
	color: #PrimaryHex;
}

.bh-nav-home:hover {
	background-color: rgba(0, 0, 0, 0.05);
}

.bh-nav-logo {
	width: 20px;
	height: 20px;
	margin-right: 5px;
}

.bh-nav-right {
	margin-right: 5%;
	display: inline-flex;
	width: fit-content;
	float: right;
}

.bh-nav-right ul {
```


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```
        list-style: none;
}

.bh-nav-right ul li {
        display: inline;
}

.bh-nav-link {
        margin: 6px;
        padding: 10px 8px;
        display: inline-block;
        font-size: 0.9em;
        text-decoration: none;
        border-radius: 8px;
        color: #SecondaryHex;
}

.bh-nav-link:hover {
        background-color: rgba(0, 0, 0, 0.1);
}
```

b. Save all changes.
c. Uncomment the "`<!-- <NavBar /> -->`" element in all "`pages/*.vue`" files.
d. Run locally: `corepack yarn dev -o`
e. Browse the website links in the browser.
f. Verify that the navigation bar is displayed correctly in all pages.
g. In **Command Prompt**, push the changes with the commit message:

Command Prompt

```
git add pages
git add components
git add "public/custom.css"
git commit -m " [UI] Add NavBar component with design and usage"
git push
```

6. Create and edit the "`components/ Copyright.vue`" file.

Copyright.vue

```
<script setup lang="ts">
  const year = new Date().getFullYear();
</script>

<template>
  <client-only>
    <footer class="bh-footer-area">
      <nav class="bh-footer-links">
          <a href="/terms">Terms of Service</a>
          <span> | </span>
          <a href="/policy">Privacy Policy</a>
          <span> | </span>
          <a href="support">Support</a>
      </nav>
```


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```
      <div class="bh-footer-copyright">
        <span>&copy; </span>
        <span>{{ year }}</span>
        <span> Company Name. All rights reserved.</span>
      </div>
    </footer>
  </client-only>
</template>
```

c. Save all changes.

7. Open the "public/custom.css" file.
   a. Append the following CSS rule sets under the `/* copyright */` comment.

custom.css

```
/* copyright */

.bh-footer-area {
      width: 100%;
      padding: 10px 0;
      font-size: 0.8em;
      text-align: center;
      color: #666;
}

.bh-footer-links a {
      font-weight: bold;
      text-decoration: none;
      color: #666;
}

.bh-footer-copyright {
      margin-top: 5px;
}
```

   b. Save all changes.
   c. Uncomment the "`<!-- <Copyright /> -->`" element in all "`pages/*.vue`" files.
   d. Browse the website links in the browser.
   e. Verify that the copyright is displayed correctly in all pages.
   f. In **Command Prompt**, push the changes.
8. Open the "`components/Cookies.vue`" file.

Cookies.vue

```
<template>
  <client-only>
    <div id="bhCookieModal" class="bh-cookie-modal bh-hidden"></div>
    <div id="bhCookieBox" class="bh-cookie-box bh-hidden">
    <span id="bhCookieX" class="bh-cookie-x">×</span>
    <div class="bh-cookie-header" >Cookies and Privacy</div>
    <div class="bh-cookie-message">
      We use cookies to ensure you get the best experience on our website.
    </div>
    <div class="bh-cookie-actions">
      <a href="/policy" class="bh-cookie-link" target="_blank">
```


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```
        <div class="bh-cookie-action bh-cookie-more">Read more</div>
      </a>
      <div id="bhCookieOK" class="bh-cookie-action bh-cookie-ok">Accept all</div>
    </div>
  </div>
  </client-only>
</template>
```

d. Save all changes.

8. Open the "public/custom.css" file.
   a. Append the following CSS rule sets under the `/* main content */` comment.

custom.css

```
/* main content */

.bh-main {
      min-height: 800px;
}

.bh-hidden {
      display: none !important;
}

.bh-cookie-modal {
      position: fixed;
      left: 0;
      top: 0;
      height: 100%;
      width: 100%;
      display: flex;
      align-items: center;
      justify-content: center;
      -webkit-backdrop-filter: blur(5px);
      backdrop-filter: blur(5px);
      background: #00000080;
      z-index: 8000;
}

.bh-cookie-box {
      position: fixed;
      left: 0;
      bottom: 0;
      margin: 30px;
      padding: 20px;
      width: 400px;
      max-width: Calc(100% - 60px);
      font-size: 0.9em;
      background-color: #fff;
      box-shadow: 0 0 10px rgba(0, 0, 0, 0.3);
      border: 1px solid rgba(0, 0, 0, 0.1);
      border-radius: 7px;
      z-index: 9000;
}
```


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```
.bh-cookie-x {
	position: absolute;
	right: 0;
	margin: 0 20px;
	line-height: 1.5em;
	cursor: pointer;
}

.bh-cookie-x:hover {
	text-shadow: 0 0 6px rgba(0, 0, 0, 0.3);
}

.bh-cookie-header {
	font-weight: bold;
	line-height: 1.5em;
}

.bh-cookie-actions {
	margin: 25px 0 8px;
	text-align: right;
}

.bh-cookie-action {
	padding: 10px 15px;
	width: 100px;
	display: inline;
	font-size: .9em;
	border-radius: 3px;
	cursor: pointer;
}

.bh-cookie-link {
	text-decoration: none;
}

.bh-cookie-more {
	color: #3F7D58;
}

.bh-cookie-more:hover {
	margin: 0 0 0 5px;
	text-decoration: underline;
}

.bh-cookie-ok {
	margin: 0 0 0 5px;
	color: #fff;
	background-color: #SecondaryColor;
}

.bh-cookie-ok:hover {
	margin: 0 0 0 5px;
	color: #fff;
	background-color: #SecondaryColorcc; /* append cc for opacity */
```


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```
}
```

a. Save all changes.
b. Uncomment the "`<!-- <Cookies /> -->`" element in the "`pages/index.vue`" file.
c. Browse the website links in the browser.
d. Verify that the cookie modal popup is displayed correctly in the home page.
e. In **Command Prompt**, push the changes.

Cookies

In web development, the client must not only able to communicate with the server seamlessly, but must also remember snippets of information for analytics, marketing, personalization, session management, site performance, state management, and multi-site integrations. The most typical storage options include cookies, LocalStorage, SessionStorage, and IndexDB. Each of them has its use cases.

Cookies were inspired by the "magic cookie" concept used for data exchanged between a browser and a web server without modification. It as a rather small data storage, up to 4KB, which is accessible by both the client (using JavaScript) and the server (on HTTP requests)—meaning all cookies created within the same domain are sent automatically sent to the server for each new HTTP requests.

1. Open the "`components/index.vue`" file.
2. Locate the "`<script setup lang="ts"></script>`" element.
3. Append the snippet to the existing script.

index.vue

```
const globalDelay = 500;

const allowCookies = useCookie<boolean>("allowCookies", {
  sameSite: "none",
  secure: true,
  maxAge: 60 * 60 * 24,
});
allowCookies.value = allowCookies.value || false;

async function loadModal() {
  const cookieModal = document.getElementById("bhCookieModal") as HTMLDivElement;
  const cookieBox = document.getElementById("bhCookieBox") as HTMLDivElement;
  const cookieX = document.getElementById("bhCookieX") as HTMLDivElement;
  const cookieOK = document.getElementById("bhCookieOK") as HTMLDivElement;
  if ( !cookieModal || !cookieBox || !cookieX || !cookieOK ) return;

  if (allowCookies.value === true) {
    allowAllCookies();
    return;
  }

  showCookiePopup();
```


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```
  cookieX.addEventListener("click", (e) => {
    showCookiePopup(false);
  });

  cookieOK.addEventListener("click", (e) => {
    allowAllCookies();
  });
}

async function showCookiePopup(show: boolean = true) {
  const cookieBox = document.getElementById("bhCookieBox") as HTMLDivElement;
  const cookieModal = document.getElementById("bhCookieModal") as HTMLDivElement;
  if (!cookieBox || !cookieModal) {
    setTimeout(showCookiePopup, 50);
    return;
  }

  if (!show) {
    cookieBox.classList.add("bh-hidden");
    cookieModal.classList.add("bh-hidden");
    return;
  }

  cookieBox.classList.remove("bh-hidden");
  cookieModal.classList.remove("bh-hidden");
}

let lastCookieClicked = 0;
async function allowAllCookies() {
  if (lastCookieClicked >= Date.now() - globalDelay) return;
  lastCookieClicked = Date.now();
  allowCookies.value = true;
  showCookiePopup(false);
}

onMounted(() => {
  setTimeout(() => {
    loadModal();
  }, 1);
});

</script>
```

a. Save all changes.
b. Browse the website links in the browser.
c. Verify that the cookie modal popup is displayed correctly in the home page.
d. Verify that if the “X” is clicked, the modal and popup is hidden.
e. Verify that if the browser is refreshed, the cookie modal and popup is shown.
f. Verify that if the “Read more” link is clicked, a new “Privacy Policy” tab is opened.
g. Verify that if the “Accept all” button is clicked, the modal and popup is hidden.
h. Verify that if the browser is refreshed, the cookie modal and popup is no longer shown.

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i. In **Command Prompt**, push the changes.

Test Website

By this time, you have unlocked the "web developer" badge. Engage in more pet projects so you could begin mastering git, HTML5, CSS3, JavaScript, TypeScript, web frameworks, and cloud platforms. Like all developers, we start with diminutive set of knowledge and skills, but it through practice that will eventually show the big difference. In your next endeavors, try using CSS frameworks and don't be shy to use ChatGPT or similar services as your coding buddy to support your development journey—just ensure that you are using it as a learning tool and not a replacement of yourself.

1. Launch a web browser.
2. Open the development link of the website.
3. **Screenshot the home page of the website.**
4. Explore the website for bugs.
   a. Fix bugs or other unusual behavior if found.
   b. Push all bug fixes.
5. Once done, submit it to your instructor via email.

To: **rodney.maniego.jr@vsu.edu.ph**
Subject: [2025-mm-dd] CSci 23 LT5 Surname

Sir Rodney,

Good day!

I am Juan D. Cruz (student #) of your CSci 23 class. Attached is the screenshot for your reference. Below is the link to my GitHub repository and the website where I contributed to.

https://github.com/username/codename
https://codename.vercel.app

Best regards,

Juan D. Cruz
BSCS-2

6. Ensure proper submission for grading; confirm with your instructor if needed.
7. Delete all your files and empty the **Recycle Bin**. (Skip if your own PC)
8. Logout your GitHub, Gmail, Vercel, and VSUEE accounts. (Skip if your own PC)
9. Close the browser, Command Prompt, and other windows.
10. Shutdown the PC and make sure the AVR/UPS is turned off.
11. Arrange the chairs before leaving.

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